Relating Fractions and Decimals

A fraction represents a part of a whole. Decimals are a kind of fraction because they show parts of a whole. For example, fifty cents ($.50) is a fraction of one dollar ($\frac{1}{2}$ dollar).

Decimals use place value to show the value of a digit. The decimal 0.7 means *seven tenths* because the digit 7 is in the tenths place. You can also write *seven tenths* as $\frac{7}{10}$.



The shaded portion of this figure can be written 0.7 or $\frac{7}{10}$. Both are read "seven tenths."

In the fraction $\frac{7}{10}$ the top number, or numerator, tells how many parts are shaded. The bottom number, the denominator, tells how many total parts there are in the figure.

$\frac{7 \rightarrow \text{numerator}}{10 \rightarrow \text{denominator}}$

When the denominator of a fraction is 10, 100, 1,000, or another multiple of 10, you can easily change the fraction to a decimal.

Example Hide

Write $\frac{75}{1{,}000}$ as a decimal.

Step 1

Think: How many decimal places does a fraction with a denominator of 1,000 need?

It needs 3 decimal places.

0. 0 7 5 ← thousandths place

Step 2

The numerator should fill the place value named by the denominator. Use placeholder zeros if necessary.

0.075 means "seventy-five thousandths"

↑ placeholder zero

Example 2: Write $\frac{15}{100}$ as a decimal.

0. 1 5

tenths hundredths

$\frac{15}{100} = 0.15$

"fifteen hundredths"

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